5.1 BLE Onboarding Procedure

Description of the onboarding process

EzloPi Power & Play BLE onboarding is the process of an EzloPi device connecting via bluetooth low energy to a gateway or smartphone for the purpose of receiving WiFi credentials and then connecting to the WiFi by using those credentials.

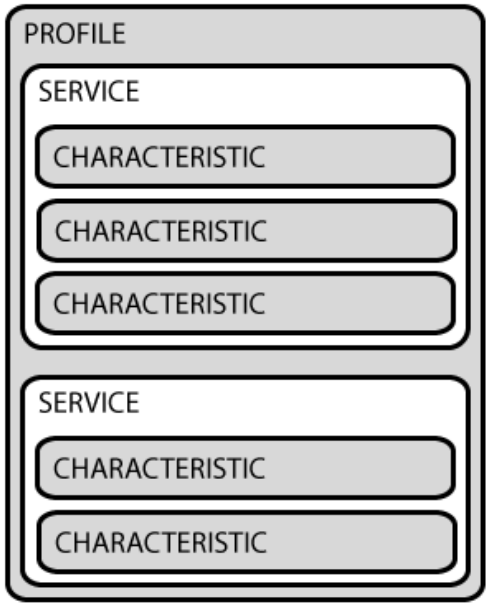
Classification of devices:

The application interacts with two types of devices: Controller and Hub/App.

1. Ezlopi Controller: is an Ezlopi device (i.e. esp32/esp32s3 based devices). It has a GATT-Service with two characteristics: 0x00FF: used to store wifi credentials, and 0x00EE used to store wifi status. The controller will be initialized as BLE-GATT-Server. The characteristics 0x00FF has the read and write permission and characteristics 0x00EE has only read permission available.

2. Hub/App:
   Hub is a device which can connect with the ezlopi-controllers. App is the android/ios/web/desktop application, which can connect with Ezlopi controllers. Hub/App will run as BLE-GATT-Client. It can connect with the Ezlopi controllers and can do some operations on it. The main operation currently is to read and write wifi-credentials and read the wifi status.

GATT transactions are used to accomplish this functionality in BLE. GATT is organized on high-level, nested objects called Profiles, Services and Characteristics, which can be seen in the illustration below:



The process starts with the EzloPi device (also known as controller) sending advertising packets periodically at a predefined advertising interval. These packets contain the name of the device which depending if the device is provisioned or not (factory) will have a different format:

- Device provisioned is formatted as ezlopi_xxxxxxxxx (where xxxxxxxxx is serial number which appears on the app).
- Device not provisioned (factory/manufacturing scenario) is formatted as ezlopi_xxxxxxxxxxxx (where xxxxxxxxxxxx is device mac address in hex).

The hub or the smartphone app (central role) will continuously scan for these advertising packets and when one is detected, it will first check if it has already been onboarded by checking its MAC address (i.e. if it's part of an internal list of connected devices). If it's a new device, it will initiate a connection to the Ezlopi device starting a standard bluetooth passkey pairing process. This part is governed by the Bluetooth GAP layer which deals with access. (GAP = Generic Access Profile). The 8-digit passkey will be hardcoded in both the hub and the Ezlopi firmware.

Once passkey is validated and pairing is complete, Hub will add the Ezlopi device mac address to an internal list of paired devices. Next the client (in our case app/hub) will send a connection request to the server (in our case Ezlopi).

Note: Ezlopi will stop advertising over BLE once a successful connection has been established which can be described as:

- For not paired devices: pairing request > paired success > connection request > connection success.
- For paired devices: search the discovered device in paired list > connection request > connection success.

Next, the hub/app(client) will check the available services and characteristics on the newly connected Ezlopi device(Server). If it finds the two required servi ces 0x00FF (characteristic 0xFF01): used to store wifi credentials, and 0x00EE service used to store wifi status it continues by writing the wifi credentials to characteristic 0xEE01. Additionally ezlopi should make available a few other characteristics (example 0xEE02, 0xEE03, 0xEE04) for writing stuff like: magic number, protocol version, owner account id.

If the Ezlopi device has not been previously paired it checks if the received credentials are new and if they are new it tries to connect to WiFi using the credentials. The result of the connection success/fail is reported by writing to the 0x00EE service and if it's a success it stores the new WiFi credentials in the flash memory as well as the user id of the owner account. If the Ezlopi device already contains an owner account user id, it will check if the hub/app (client) sending new wifi credentials is using the same user id and only accept these credentials from the same user id.

Note: at this phase both WiFi and BLE connections are maintained active so as a result, Hub/App can now attempt to read 0x00EE from Ezlopi to check the connection status

The Ezlopi device will now attempt to connect to wifi using the provided credentials and will report the status of the wifi connection in the 0x00E1 characteristic. Hub/App can read that register from the Ezlopi device and report success/failure to the user.

Error handling

These BLE connections are exclusive, meaning the BLE peripheral can only be connected to one central device (a mobile phone or hub etc.) at a time! So what happens when the user has both a hub and smartphone trying to scan? This is something that might occasionally happen in practice so the app/hub needs to be able to respond with a proper error message when this happens.

Since Ezlopi is based on the ESP32 chipset, it only has 2.4GHz WiFi radio capability. This means that the hub/app will have to check the WiFi station used for pairing, operates in the 2.4GHz spectrum, before using these credentials for the onboarding process.

The ESP32 (and thus Ezlopi firmware) has the capability to detect specific reasons for wifi connection failure (AP down, wrong password, timeout, etc.). Please report specific connection failure reasons (via 0x00E1 characteristic) and act accordingly in ezlopi fw and hub/app.

What happens if the WiFi network is not accessible or the connection fails? We have 3 distinct cases here:

1. Ezlopi device has been already onboarded in the past, but the wifi is down and it can't connect. For this case it will just continue to retry the connection at a predefined interval for a predefined number of retries. After that number of retries is exhausted, Ezlopi should start advertising via BLE and accept new wifi credentials from the hub/app which is using the same user id as the one which originally onboarded the Ezlopi device.
2. Ezlopi has not finished the onboarding process, it will go back to advertising via BLE, waiting for new wifi credentials, trying again and again.
3. Ezlopi has been onboarded in the past but wifi credentials changed. (i.e ezlopi reports a wrong password on wifi connection). In this case Ezlopi should switch back to ble advertising mode, but only accept new wifi credentials from the same account id).

Security Flaws:

As mentioned above, Firmware needs to have the 8 digit passkey for BLE authentication. However, different passkeys will be assigned for different models of the ezlopi. Since the means of storing the passkey into the device is not defined yet. We assume ezlopi will contain the passkey as fixed-key. Where fixed key can easily be fetched from the firmware, since the firmware is open-source. With this flaw ezlopi devices can be used as hacking tools, if not fixed.

Probable Remedies:

1. Can attach the passkey to the 'id.bin' file. So, each device will have their own unique passkey which can be fetched from the cloud whenever required, by the authentic user. 'id.bin' file is created while registering a new device, either by using "ezlogic-qt-app" or "web-flasher".
2. Can always renew the passkey after successful connection with app/hub, and app/hub can keep track of the updated passkey. If the passkey is lost then one has to reset the ezlopi-device so that one can connect using the default passkey.

BLE Operations on Controller side:

BLE Operations on Hub/App side:

Ezlopi device association with customer